

1. The Core Workflow: Real-Time Disruption & Autonomous Rebooking

This is the absolute heart of the product. It must execute flawlessly without the user lifting a finger.

- **The Trigger:** A webhook or high-frequency poll from the Amadeus/Sabre API flags a cancellation or a delay that makes a connection mathematically impossible (e.g., delay > layover time).
- **Agentic Logic:**
 - **State Check:** Assess the current location of the user and the final destination.
 - **Option Generation:** Query the flight API for the next 3 available routes.
 - **Policy Evaluation:** The agent cross-references the cost of these flights against the credit card's built-in travel insurance or policy limits.
 - **Hotel Check (If overnight):** If the next flight is tomorrow, the agent queries hotel APIs for rooms near the airport within the allowed per diem.
- **The Action:**
 - Executes the booking payload to the airline and hotel APIs.
 - Pushes a real-time notification to the mobile app: *"Your flight was cancelled. We've automatically rebooked you on Delta 451 at 8 PM and secured a room at the Marriott. Click to view boarding passes."*

2. The Predictive "Pre-Disruption" Watcher

Don't wait for the official cancellation. A truly intelligent concierge acts on leading indicators before the rest of the airport starts scrambling.

- **The Trigger:** Weather API alerts (e.g., severe thunderstorms at the connecting hub) or inbound aircraft tracking (the plane meant to fly the user is grounded two cities away).
- **Agentic Logic:** A probabilistic model evaluates the likelihood of a missed connection. If the probability crosses a high threshold (e.g., >85%), the agent activates.
- **The Action:**
 - Places a temporary "soft hold" on seats for a backup flight (if the API allows).
 - Alerts the user: *"High risk of missed connection in Atlanta due to weather. We are holding a backup seat on the 9 PM flight just in case."*

3. The Cascading Logistics Updater (Ground Transit)

A flight change breaks everything downstream. This workflow proves you understand the end-to-end travel experience.

- **The Trigger:** The Core Workflow successfully completes a flight rebooking.
- **Agentic Logic:** The agent scans the user's connected itinerary for downstream dependencies—rental cars, airport transfers, or hotel check-ins at the final destination.
- **The Action:**
 - Calls the Hertz/Enterprise API to update the rental car pickup time.
 - Modifies the hotel check-in date if the user is arriving a day late.
 - Sends a summary card to the app showing updated ground logistics.

4. The Human-in-the-Loop Conversational Fallback

Even with perfect logic, users sometimes want control. If the autonomous choice isn't their preference, they need a frictionless way to change it.

- **The Trigger:** The user taps "View Options" on the automated rebooking notification in the app.
- **Agentic Logic:** An LLM-powered chat interface (e.g., using LangChain) loads the user's context. The user types: *"Actually, I'm exhausted. Can I just fly out tomorrow morning instead?"*
- **The Action:**
 - The agent parses the intent, unwinds the current autonomous booking, queries morning flights, and presents a 9 AM option.
 - Executes the change upon user confirmation via a one-tap button.

5. The Automated Claim & Compensation Filer

Travel disruptions often entitle passengers to money (like EU261 compensation) or trigger credit card insurance payouts, but nobody wants to fill out the paperwork.

- **The Trigger:** The disrupted trip concludes, and the user arrives at their final destination.
- **Agentic Logic:** The agent calculates total delay time and maps it against airline regulations or the card member's travel insurance policy limits.
- **The Action:**
 - Automatically generates the PDF claim forms using the user's booking data.
 - Submits the claim to the airline via email/API or directly to the credit card's insurance portal.
 - Notifies the user: *"We've filed a \$400 delay compensation claim on your behalf."*